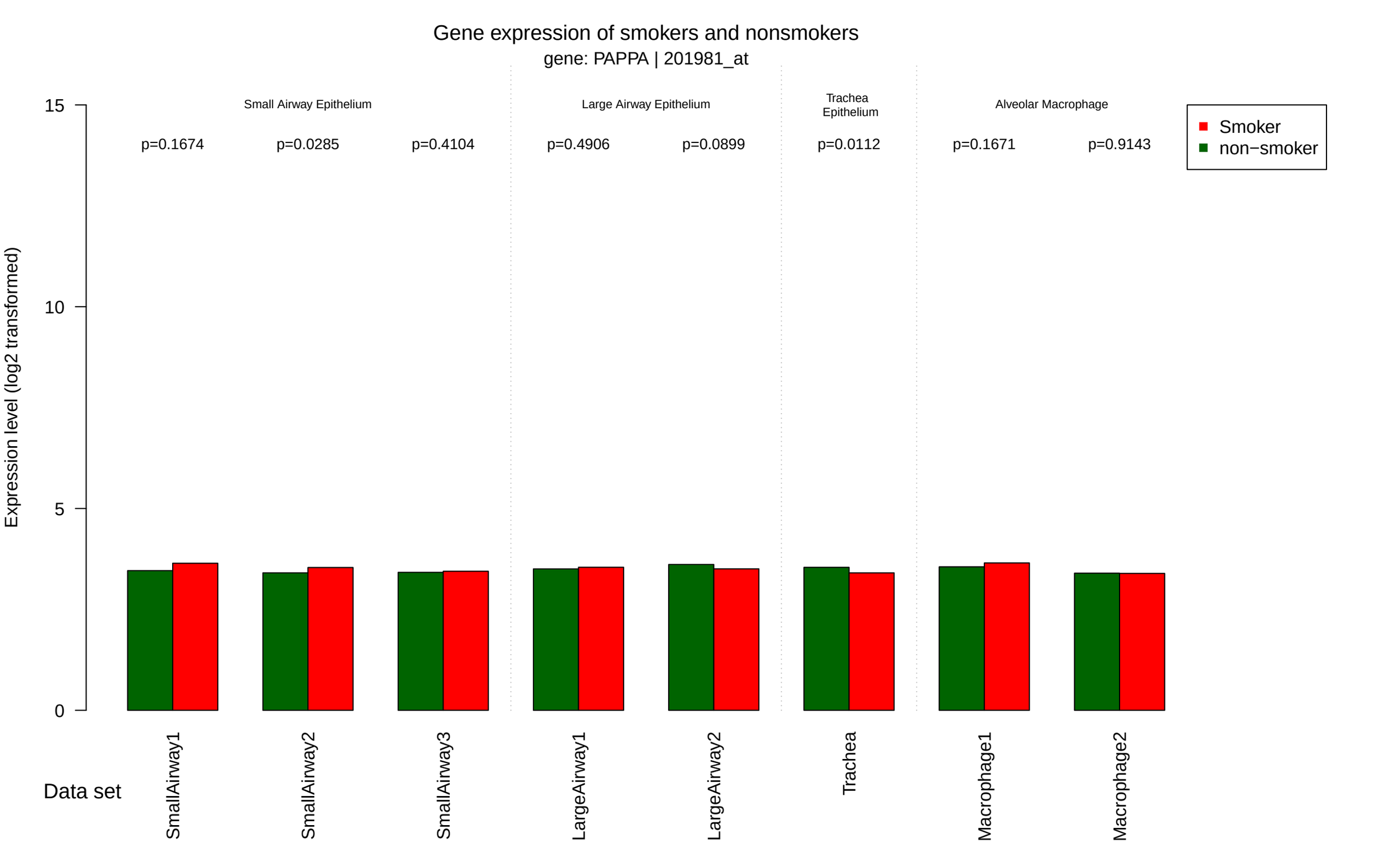
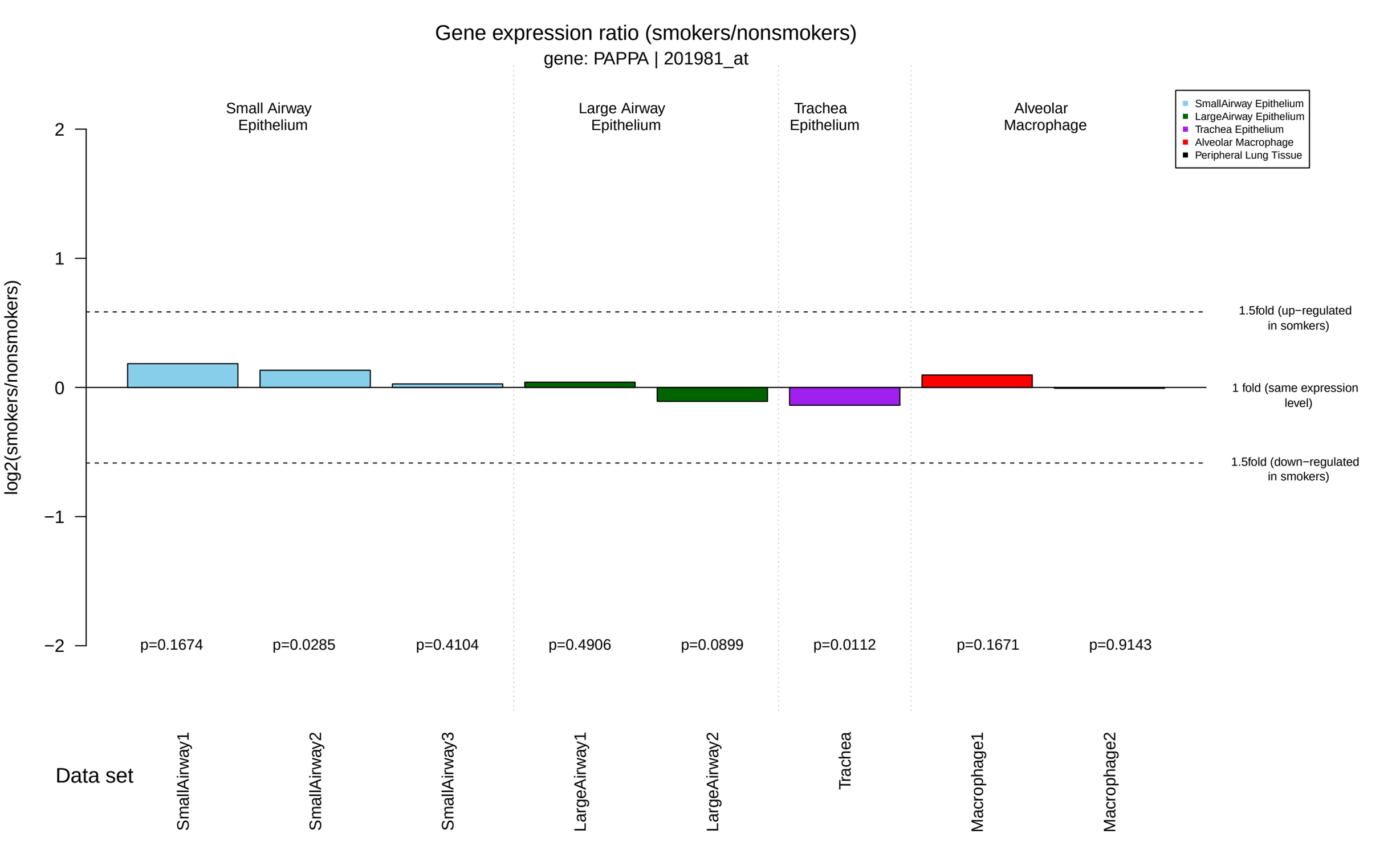
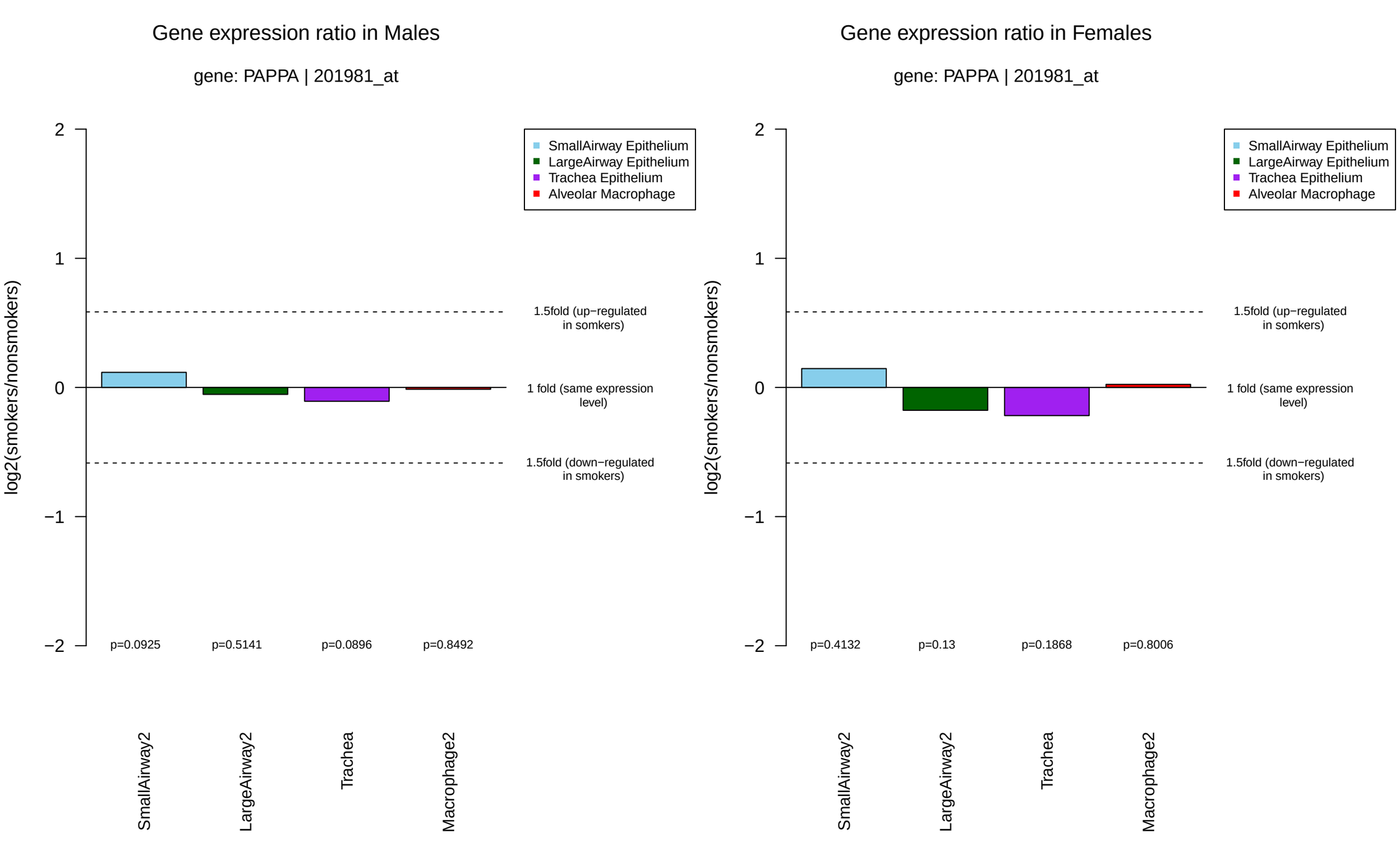
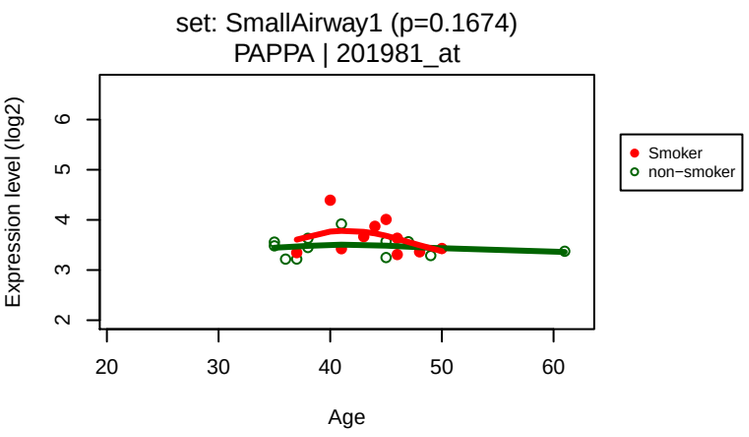
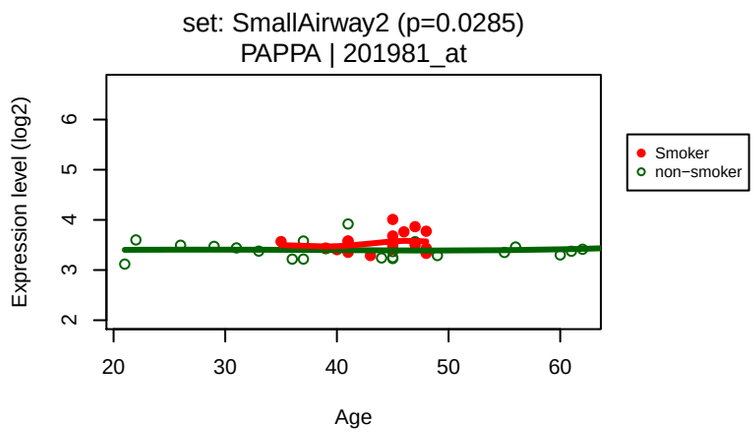
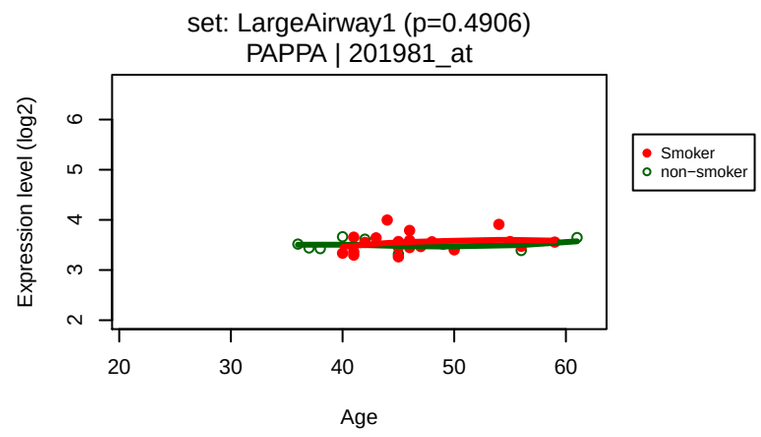

Gene expression of smokers and nonsmokers
gene: PAPPA | 201981_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1674
p=0.0285
p=0.4104
p=0.4906
p=0.0899
p=0.0112
p=0.1671
p=0.9143
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

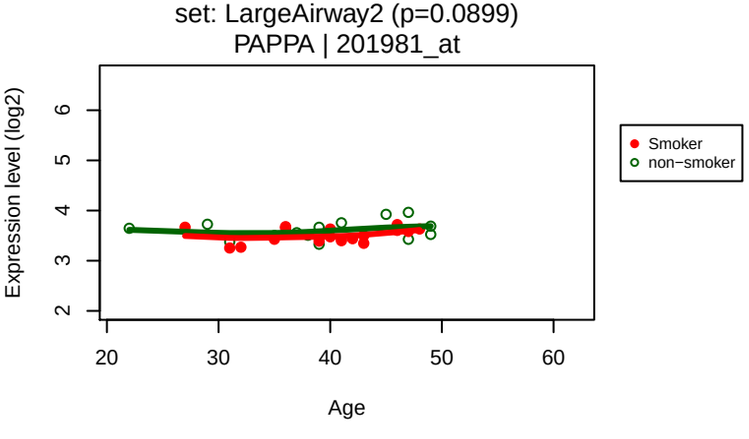

Gene expression ratio (smokers/nonsmokers)
gene: PAPPA | 201981_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1674
p=0.0285
p=0.4104
p=0.4906
p=0.0899
p=0.0112
p=0.1671
p=0.9143
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

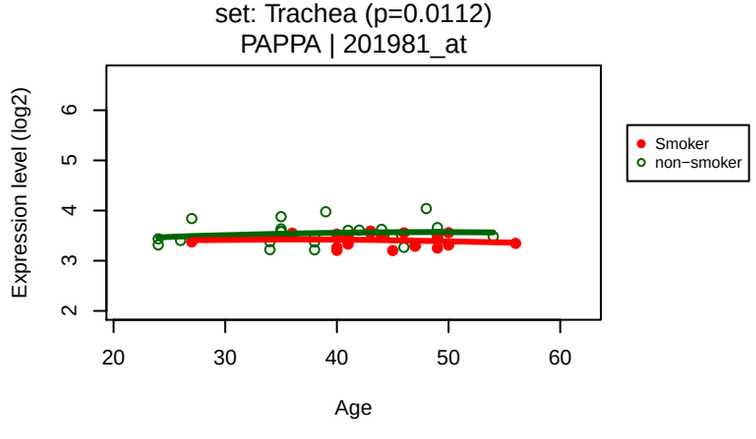

Gene expression ratio in Males
gene: PAPPA | 201981_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0925
p=0.5141
p=0.0896
p=0.8492
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PAPPA | 201981_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4132
p=0.13
p=0.1868
p=0.8006
SmallAirway2
LargeAirway2
Trachea
Macrophage2

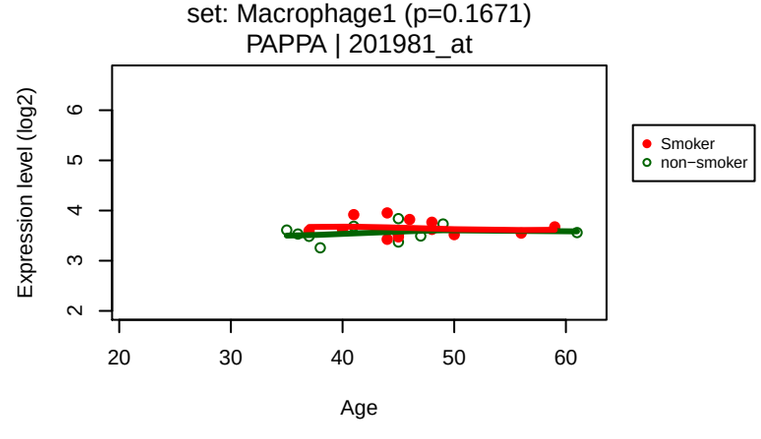

set: SmallAirway1 (p=0.1674)
PAPPA | 201981_at
Expression level (log2)
Age
Smoker
non−smoker

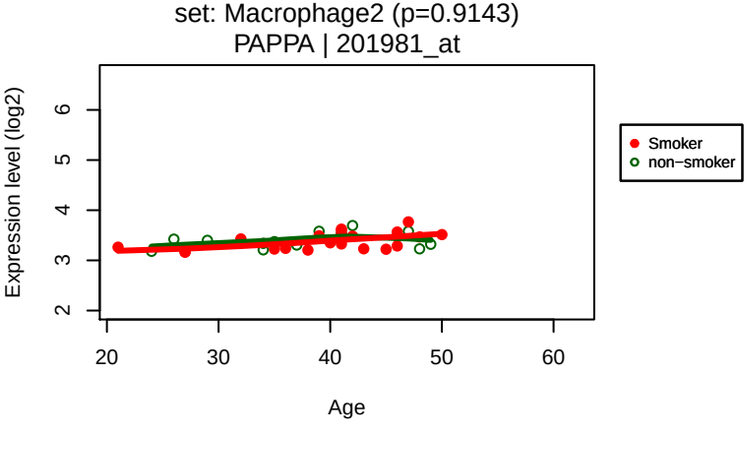

set: SmallAirway2 (p=0.0285)
PAPPA | 201981_at
Expression level (log2)
Age
Smoker
non−smoker

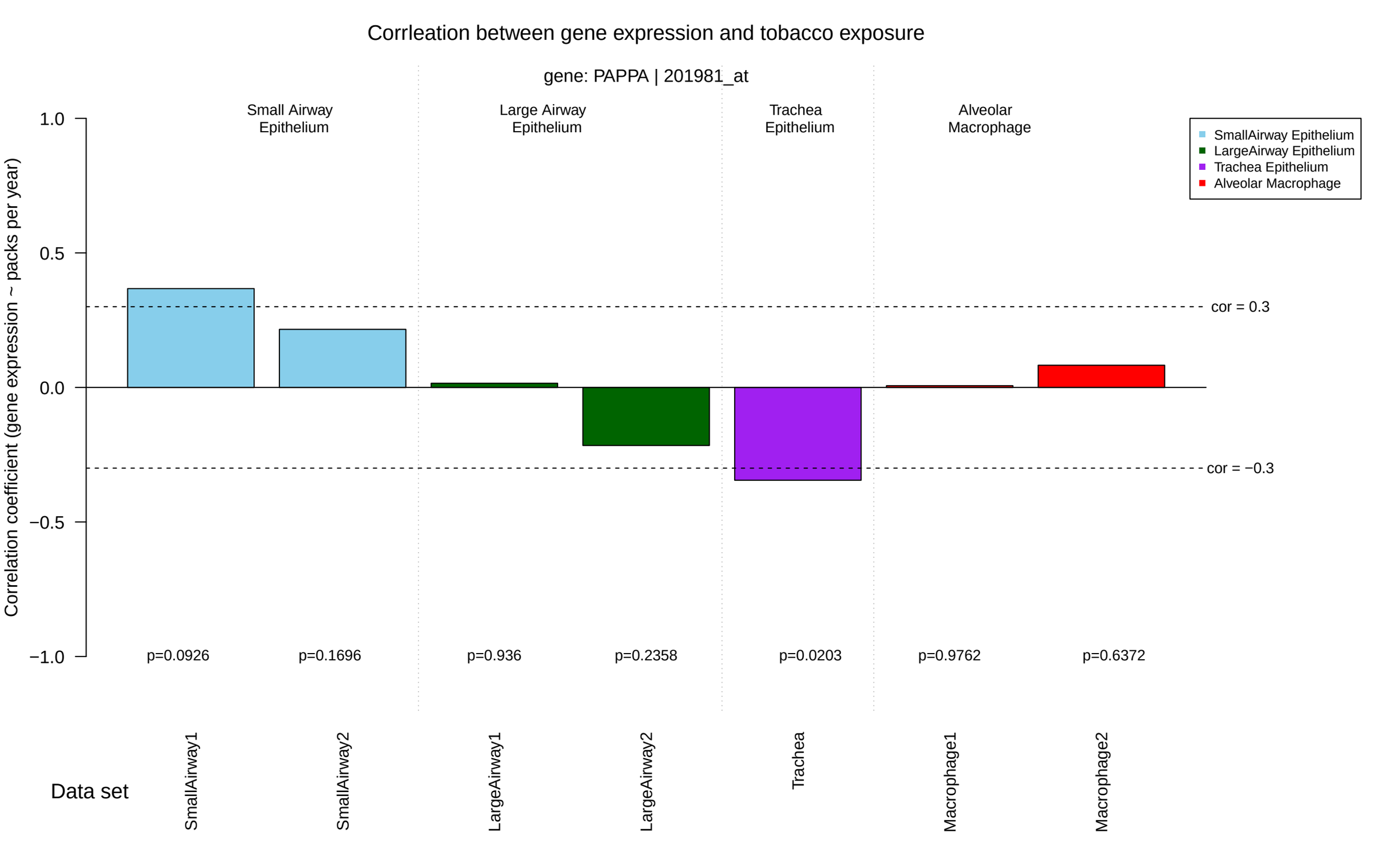

set: LargeAirway1 (p=0.4906)
PAPPA | 201981_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0899)
PAPPA | 201981_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0112)
PAPPA | 201981_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1671)
PAPPA | 201981_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9143)
PAPPA | 201981_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PAPPA | 201981_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0926
p=0.1696
p=0.936
p=0.2358
p=0.0203
p=0.9762
p=0.6372
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2